

07/09/26

Signal Processing

Recap: Demo; $x[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/N}$; $X[k,j] = \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} x[m,n] e^{j2\pi km/N} e^{-j2\pi jn/N}$

* The importance of phase *

$$|x[k,j]|, \angle x[k,j]$$

DFT - Eigen function of LTI system

- Symmetry properties: $x^*[n]$, $x^*[-n]$, $\text{Re}\{x[n]\}$, $j\text{Im}\{x[n]\}$, real signals.

- Theorems: Linearity, time shifting, frequency shifting, time reversal, convolution

The importance of phase.

Analysis equations: $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi nk/N}$; $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$

Synthesis equations: $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}$; $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$

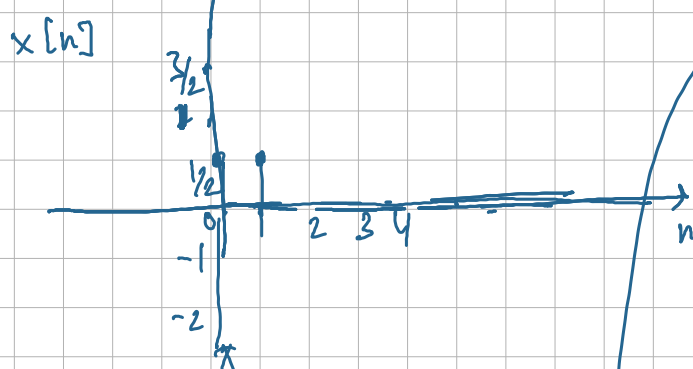
DFT spectrum DTFT-spectrum

Observation: $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} |x[k]| e^{j\angle X(k)} e^{j2\pi kn/N}$

mag phase

$$\neq \frac{1}{N} \sum_{k=0}^{N-1} |x[k]| e^{j2\pi kn/N}$$

Illustration:



$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$X(e^{j\omega}) = \frac{1}{2} (1 + e^{-j\omega})$$

$$X(e^{j\omega}) = \cos\left(\frac{\omega}{2}\right) e^{-j\frac{\omega}{2}}$$

$$|X(e^{j\omega})| = \left| \cos\left(\frac{\omega}{2}\right) \right|$$

$$\angle X(e^{j\omega}) = -\frac{\omega}{2}$$

If $\angle X(e^{j\omega})$ is forcibly set to 0, we cannot recover $x[n]$

For some $x[n]$ which is absolutely summable, i.e., the DTFT exists, $X(e^{j\omega})$ is its DTFT. $X(e^{j\omega})$ is the spectrum of the signal $x[n]$.

Let's pick an arbitrary frequency ω_0 . $X(e^{j\omega_0})$ gives us the spectral component of $x[n]$ at $\omega = \omega_0$

$$X(e^{j\omega_0}) = |X(e^{j\omega_0})| \cdot e^{j \angle X(e^{j\omega_0})}$$

This component is multiplied with $e^{j\omega_0 n}$ during reconstruction.

i.e.

$$\underbrace{|X(e^{j\omega_0})| \cdot e^{j \angle X(e^{j\omega_0})}}_{\text{cos } \omega_0 n} \cdot (\cos \omega_0 n + j \sin \omega_0 n)$$

$$= |X(e^{j\omega_0})| (\cos(\omega_0 n + \angle X(e^{j\omega_0})) + j \sin(\omega_0 n + \angle X(e^{j\omega_0})))$$

$$h[n] = \begin{cases} \frac{1}{M_2 - M_1 + 1} & M_1 \leq n \leq M_2 \\ 0 & \text{else} \end{cases} \quad M_1 < 0, M_2 > 0$$

Q: Find $H(e^{j\omega})$.

$$\sum_{k=M_1}^{M_2} \alpha^k = \frac{\alpha^{M_1} - \alpha^{M_2+1}}{1-\alpha} \quad \alpha < 1$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n] e^{-j\omega n} = \frac{1}{M_2 - M_1 + 1} \sum_{n=M_1}^{M_2} e^{-j\omega n}$$

$$= k \left(\frac{e^{-j\omega M_1} - e^{-j\omega (M_2+1)}}{1 - e^{-j\omega}} \right) \quad ; \quad k = \frac{1}{M_2 - M_1 + 1}$$

$$= k \cdot \frac{e^{-j(\frac{\omega}{2})}}{e^{-j\omega/2}} \cdot \frac{\sin(\frac{\omega}{2})}{\sin \frac{\omega}{2}}$$

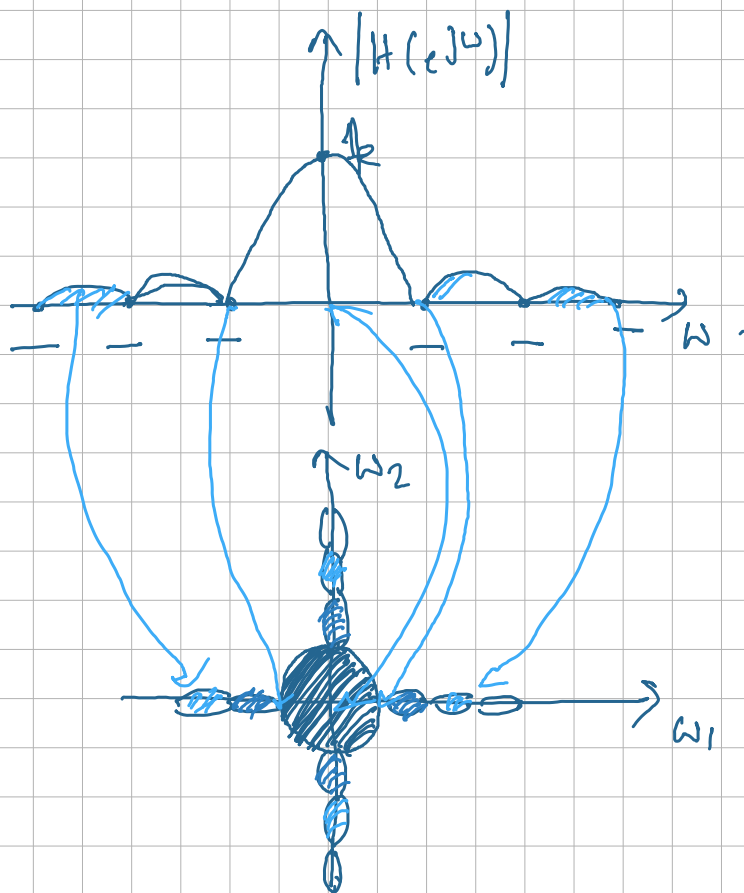
$$\text{Num: } k \cdot e^{-j\omega M_1} \left(1 - e^{-j\omega(M_2+1) + j\omega M_1} \right) = e^{-j\omega M_1} \left(1 - e^{-j\omega(M_2+1-M_1)} \right)$$

$$= e^{-j\omega M_1} \left(e^{j\frac{\omega}{2}(M_2-M_1+1)} - e^{-j\frac{\omega}{2}(M_2-M_1+1)} \right) e^{-j\frac{\omega}{2}(M_2-M_1+1)}$$

$$= e^{-j\omega M_1} \cdot e^{-j\frac{\omega}{2}(M_2-M_1+1)} \cdot 2j \sin \frac{\omega}{2} (M_2-M_1+1)$$

$$\text{Den: } e^{-j\frac{\omega}{2}} \cdot 2j \sin \frac{\omega}{2}$$

$$\therefore H(e^{j\omega}) = k \cdot \frac{e^{-j\omega M_1} e^{-j\frac{\omega}{2}(M_2-M_1+1)} \cdot \sin \frac{\omega}{2} (M_2-M_1+1)}{e^{-j\omega/2} \cdot \sin \frac{\omega}{2}}$$



- Existence of the DTFT of $x[n]$. The sufficient condition for $X(e^{j\omega})$ to exist is absolute summability of $x[n]$.

$$\begin{aligned}
 \underbrace{|X(e^{j\omega})|} &= \left| \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right| \\
 &\leq \sum_{n=-\infty}^{\infty} |x[n] e^{-j\omega n}| \\
 &= \sum_{n=-\infty}^{\infty} |x[n]| \\
 &< \infty
 \end{aligned}$$

$\therefore$ Absolute summability is a sufficient condition for the DTFT of $x[n]$ to exist.

• We assume the sufficiency condition is satisfied in the following.

- If $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$, what is the DTFT of $x^*[n]$?

$$\Rightarrow \text{DTFT}(x^*[n]) = X^*(e^{j\omega})$$